Week 4 Day 2 — Core Supervised Learning: Write-Up

Adult Census Income dataset | Preprocessing, models and evaluation | Aqsa Saqib

1. What I did

I used the same split as Day 1: 26,048 rows for training and 6,513 rows as the hold-out test set. I built two models, Logistic Regression and Decision Tree. Each one is a pipeline (preprocessing + model) and was trained on the training set only. Then I tested both on the hold-out test set.

2. Preprocessing choices

Step	What I chose	Why	Alternative I skipped
Missing numbers	Median	Not affected by the huge values in capital.gain	Mean; dropping rows
Number scale	StandardScaler	Puts all numbers on the same scale, so capital.gain does not dominate age	-
Missing text	Fill with "Missing"	"Not answered" can itself be useful information	most_frequent
Text to numbers	OneHotEncoder (handle_unknown = ignore)	Categories have no order. A new category in test data does not cause an error	Ordinal / Label encoding (creates a fake order)
Columns left out	fnlwgt, education	fnlwgt is a census weight. education is the same as education.num	-

- **Features:** 5 numeric and 7 categorical. After one-hot encoding there are 90 columns.
- **No data leakage:** preprocessing is inside the pipeline, so the median, the scaling and the categories are learned from the training data only.

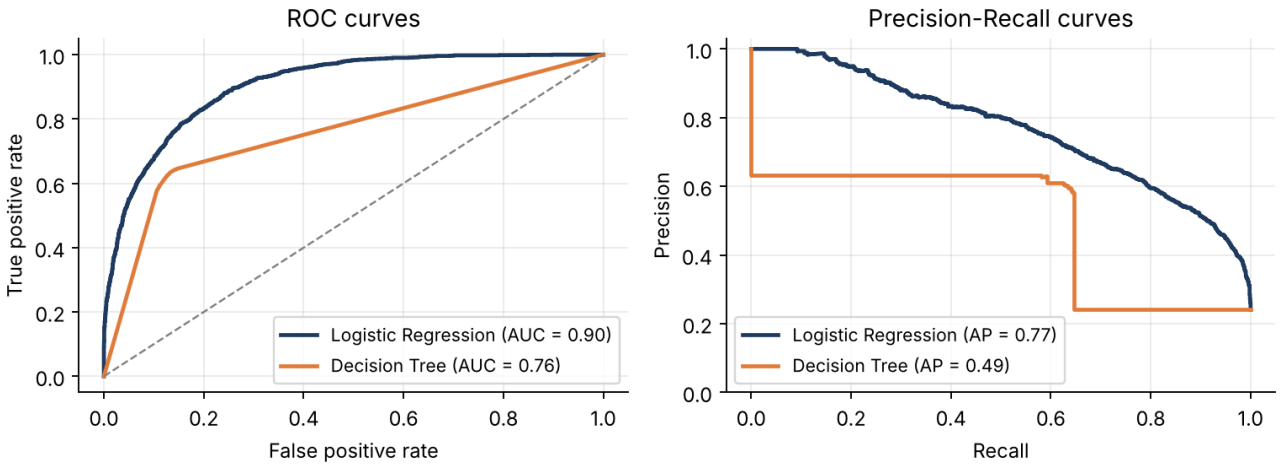
3. Model metrics (hold-out test set)

Model	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
Day 1 baseline (education.num >= 13)	0.758	0.498	0.511	0.505	0.674	0.372
Logistic Regression	**0.853**	**0.735**	**0.610**	**0.667**	**0.904**	**0.767**
Decision Tree	0.817	0.628	0.592	0.610	0.758	0.493

Confusion matrix	True negative	False positive	False negative	True positive
Logistic Regression	4,600	345	611	957
Decision Tree	4,394	551	639	929

- Both models beat the Day 1 baseline. Logistic Regression is the best on every metric.
- In both models, false negatives (missed high earners) are more common than false positives. So recall is lower than precision, and recall is what holds the F1 score down.

ROC and Precision-Recall curves (hold-out test set)



The Logistic Regression curve is above the Decision Tree curve in both charts, so it is the better model.

4. Looking inside the models

- **Logistic Regression, more likely >50K:** capital gain, married people, more education, manager jobs.
- **Logistic Regression, more likely <=50K:** house service jobs, young people living with parents, never-married people, female.
- Some country values are large but come from very few people, so they are not reliable. Female can show bias in the data.
- **Decision Tree overfits:** it is 48 levels deep. Train accuracy is 0.979 but test accuracy is 0.817, so it memorised the training data.
- **Logistic Regression does not overfit:** train accuracy 0.852, test accuracy 0.853.
- **Top 3 splits of the tree:** (1) Is the person married? (2) If no, is the capital gain high? (3) If yes, is the education high (Bachelors or more)? This makes sense and matches what I saw on Day 1.

5. Which model I will continue with

I will continue with **Logistic Regression**. It has the best F1 score (0.667). The Decision Tree has 0.610 and the Day 1 baseline has 0.505. It also has the best ROC AUC (0.904 against 0.758). It does not overfit, and it is easy to explain.

I will also try the **Decision Tree** again, but smaller. Right now it is too deep and it overfits.